I²C Communication Protocol

What is I²C (Inter-Integrated Circuit)?

I²C (Inter-Integrated Circuit) is a serial communication protocol used to connect a microcontroller with multiple devices using two communication lines.

Communication Lines

I²C uses two main communication lines:

- SDA (Serial Data Line) – Transfers data between devices.
- SCL (Serial Clock Line) – Provides the clock signal for communication.

What is Master and Slave ?

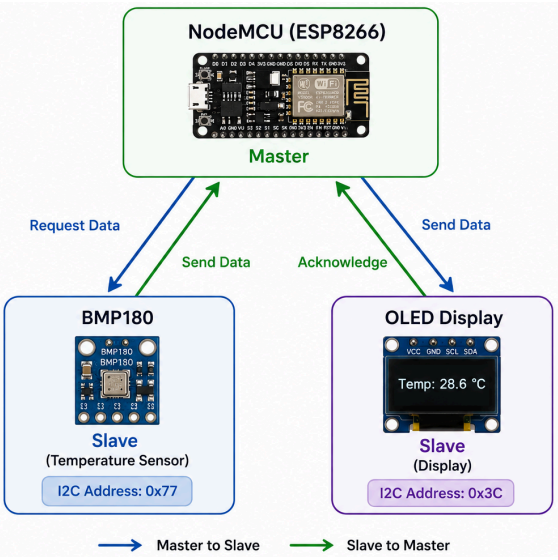
In I²C communication:

- Master – Controls the communication and starts the communication.
- Slave – Responds to the Master.
- Each Slave device has a unique I²C address.

Example: Smartwatch

A smartwatch can use multiple I²C devices:

- Microcontroller → Master
- BMP180 Temperature Sensor → Slave
- Accelerometer → Slave
- OLED Display → Slave
- Master communicates with each device using its unique I²C address.



How I²C Works

1. Master starts communication.
2. Master selects a device using its address.

3. Master sends or requests data.
4. Slave responds.
5. Data is transferred through SDA and SCL.

Features

- Uses only two communication lines.
- Supports multiple devices.
- Each device has an I²C address.
- Reduces wiring.
- Suitable for sensors and displays.

Applications

- Smartwatches
- IoT devices
- OLED displays
- Temperature sensors
- RTC modules
- EEPROM memory
- Accelerometers and gyroscopes

I²C Activity: OLED with NodeMCU

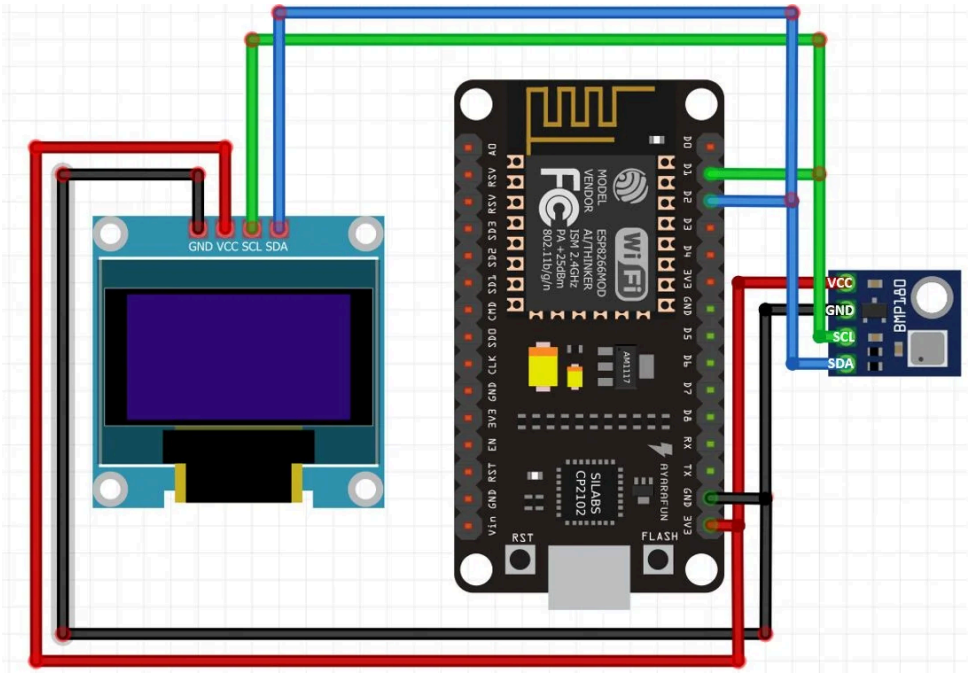
To connect an SSD1306 128×64 OLED display to NodeMCU using I²C and display text.

Components

- NodeMCU ESP8266
- SSD1306 I²C OLED
- Jumper wires
- USB cable
- Computer with Arduino IDE

Connections

OLED Display	BMP180	NodeMCU
VCC	VCC	3.3V
GND	GND	GND
SDA	SDA	D2
SCL	SCL	D1



Required Libraries

- Libraries Required
- Adafruit SSD1306
- Adafruit GFX Library
- Adafruit BMP085 Library

Code:

Expected Output

HELLO